MSG No. 191.234 $P6/mmm1'$ [Type II, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `6/mmm1'`

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]

Table 2: Wyckoff site: `1b`, site symmetry: `6/mmm1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]

Table 3: Wyckoff site: `2c`, site symmetry: `-6m21'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,3,5,10,11,12,14,16,18,19,20,21,25,27,29,34,35,36,38,40,42,43,44,45]
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[2,4,6,7,8,9,13,15,17,22,23,24,26,28,30,31,32,33,37,39,41,46,47,48]

Table 4: Wyckoff site: `2d`, site symmetry: `-6m21'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,3,5,10,11,12,14,16,18,19,20,21,25,27,29,34,35,36,38,40,42,43,44,45]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[2,4,6,7,8,9,13,15,17,22,23,24,26,28,30,31,32,33,37,39,41,46,47,48]

Table 5: Wyckoff site: `2e`, site symmetry: `6mm1'`

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,4,5,6,19,20,21,22,23,24,25,26,27,28,29,30,43,44,45,46,47,48]
2	$[0, 0, -z]$	[7,8,9,10,11,12,13,14,15,16,17,18,31,32,33,34,35,36,37,38,39,40,41,42]

Table 6: Wyckoff site: `3f`, site symmetry: `mmm1'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,7,11,13,16,19,23,25,28,31,35,37,40,43,47]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[2,5,9,10,14,17,21,22,26,29,33,34,38,41,45,46]
3	$\left[0, \frac{1}{2}, 0\right]$	[3,6,8,12,15,18,20,24,27,30,32,36,39,42,44,48]

Table 7: Wyckoff site: 3g, site symmetry: mmm1'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,4,7,11,13,16,19,23,25,28,31,35,37,40,43,47]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,5,9,10,14,17,21,22,26,29,33,34,38,41,45,46]
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[3,6,8,12,15,18,20,24,27,30,32,36,39,42,44,48]

Table 8: Wyckoff site: 4h, site symmetry: 3m.1'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5,19,20,21,25,27,29,43,44,45]
2	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[2,4,6,22,23,24,26,28,30,46,47,48]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,8,9,13,15,17,31,32,33,37,39,41]
4	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[10,11,12,14,16,18,34,35,36,38,40,42]

Table 9: Wyckoff site: 6i, site symmetry: 2mm1'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	[1,4,19,23,25,28,43,47]
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[2,5,21,22,26,29,45,46]
3	$\left[0, \frac{1}{2}, z\right]$	[3,6,20,24,27,30,44,48]
4	$\left[\frac{1}{2}, 0, -z\right]$	[7,11,13,16,31,35,37,40]
5	$\left[0, \frac{1}{2}, -z\right]$	[8,12,15,18,32,36,39,42]
6	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[9,10,14,17,33,34,38,41]

Table 10: Wyckoff site: 6j, site symmetry: m2m1'

No.	position	mapping
1	$[x, 0, 0]$	[1,7,16,23,25,31,40,47]
2	$[x, x, 0]$	[2,10,17,21,26,34,41,45]
3	$[0, x, 0]$	[3,8,18,24,27,32,42,48]
4	$[-x, 0, 0]$	[4,11,13,19,28,35,37,43]
5	$[-x, -x, 0]$	[5,9,14,22,29,33,38,46]
6	$[0, -x, 0]$	[6,12,15,20,30,36,39,44]

Table 11: Wyckoff site: 6k, site symmetry: m2m1'

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	[1,7,16,23,25,31,40,47]
2	$\left[x, x, \frac{1}{2}\right]$	[2,10,17,21,26,34,41,45]
3	$\left[0, x, \frac{1}{2}\right]$	[3,8,18,24,27,32,42,48]
4	$\left[-x, 0, \frac{1}{2}\right]$	[4,11,13,19,28,35,37,43]
5	$\left[-x, -x, \frac{1}{2}\right]$	[5,9,14,22,29,33,38,46]
6	$\left[0, -x, \frac{1}{2}\right]$	[6,12,15,20,30,36,39,44]

Table 12: Wyckoff site: `6l`, site symmetry: `mm21'`

No.	position	mapping
1	$[x,\ 2x,\ 0]$	`[1,11,16,19,25,35,40,43]`
2	$[-x,\ x,\ 0]$	`[2,9,17,22,26,33,41,46]`
3	$[-2x,\ -x,\ 0]$	`[3,12,18,20,27,36,42,44]`
4	$[-x,\ -2x,\ 0]$	`[4,7,13,23,28,31,37,47]`
5	$[x,\ -x,\ 0]$	`[5,10,14,21,29,34,38,45]`
6	$[2x,\ x,\ 0]$	`[6,8,15,24,30,32,39,48]`

Table 13: Wyckoff site: `6m`, site symmetry: `mm21'`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{2}\right]$	`[1,11,16,19,25,35,40,43]`
2	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[2,9,17,22,26,33,41,46]`
3	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	`[3,12,18,20,27,36,42,44]`
4	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	`[4,7,13,23,28,31,37,47]`
5	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[5,10,14,21,29,34,38,45]`
6	$\left[2x,\ x,\ \frac{1}{2}\right]$	`[6,8,15,24,30,32,39,48]`

Table 14: Wyckoff site: `12n`, site symmetry: `..m1'`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,23,25,47]`
2	$[x,\ x,\ z]$	`[2,21,26,45]`
3	$[0,\ x,\ z]$	`[3,24,27,48]`
4	$[-x,\ 0,\ z]$	`[4,19,28,43]`
5	$[-x,\ -x,\ z]$	`[5,22,29,46]`
6	$[0,\ -x,\ z]$	`[6,20,30,44]`
7	$[x,\ 0,\ -z]$	`[7,16,31,40]`
8	$[0,\ x,\ -z]$	`[8,18,32,42]`
9	$[-x,\ -x,\ -z]$	`[9,14,33,38]`
10	$[x,\ x,\ -z]$	`[10,17,34,41]`
11	$[-x,\ 0,\ -z]$	`[11,13,35,37]`
12	$[0,\ -x,\ -z]$	`[12,15,36,39]`

Table 15: Wyckoff site: `12o`, site symmetry: `.m.1'`

No.	position	mapping
1	$[x,\ 2x,\ z]$	`[1,19,25,43]`
2	$[-x,\ x,\ z]$	`[2,22,26,46]`
3	$[-2x,\ -x,\ z]$	`[3,20,27,44]`
4	$[-x,\ -2x,\ z]$	`[4,23,28,47]`
5	$[x,\ -x,\ z]$	`[5,21,29,45]`
6	$[2x,\ x,\ z]$	`[6,24,30,48]`
7	$[-x,\ -2x,\ -z]$	`[7,13,31,37]`

continued ...

Table 15

No.	position	mapping
8	$[2x,\ x,\ -z]$	`[8,15,32,39]`
9	$[-x,\ x,\ -z]$	`[9,17,33,41]`
10	$[x,\ -x,\ -z]$	`[10,14,34,38]`
11	$[x,\ 2x,\ -z]$	`[11,16,35,40]`
12	$[-2x,\ -x,\ -z]$	`[12,18,36,42]`

Table 16: Wyckoff site: `12p`, site symmetry: `m..1'`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,16,25,40]`
2	$[x-y,\ x,\ 0]$	`[2,17,26,41]`
3	$[-y,\ x-y,\ 0]$	`[3,18,27,42]`
4	$[-x,\ -y,\ 0]$	`[4,13,28,37]`
5	$[-x+y,\ -x,\ 0]$	`[5,14,29,38]`
6	$[y,\ -x+y,\ 0]$	`[6,15,30,39]`
7	$[x-y,\ -y,\ 0]$	`[7,23,31,47]`
8	$[y,\ x,\ 0]$	`[8,24,32,48]`
9	$[-x,\ -x+y,\ 0]$	`[9,22,33,46]`
10	$[x,\ x-y,\ 0]$	`[10,21,34,45]`
11	$[-x+y,\ y,\ 0]$	`[11,19,35,43]`
12	$[-y,\ -x,\ 0]$	`[12,20,36,44]`

Table 17: Wyckoff site: `12q`, site symmetry: `m..1'`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	`[1,16,25,40]`
2	$\left[x-y,\ x,\ \frac{1}{2}\right]$	`[2,17,26,41]`
3	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[3,18,27,42]`
4	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[4,13,28,37]`
5	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[5,14,29,38]`
6	$\left[y,\ -x+y,\ \frac{1}{2}\right]$	`[6,15,30,39]`
7	$\left[x-y,\ -y,\ \frac{1}{2}\right]$	`[7,23,31,47]`
8	$\left[y,\ x,\ \frac{1}{2}\right]$	`[8,24,32,48]`
9	$\left[-x,\ -x+y,\ \frac{1}{2}\right]$	`[9,22,33,46]`
10	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[10,21,34,45]`
11	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[11,19,35,43]`
12	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[12,20,36,44]`

Table 18: Wyckoff site: `24r`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,25]`
2	$[x-y,\ x,\ z]$	`[2,26]`

continued ...

Table 18

No.	position	mapping
3	$[-y,\, x-y,\, z]$	[3,27]
4	$[-x,\, -y,\, z]$	[4,28]
5	$[-x+y,\, -x,\, z]$	[5,29]
6	$[y,\, -x+y,\, z]$	[6,30]
7	$[x-y,\, -y,\, -z]$	[7,31]
8	$[y,\, x,\, -z]$	[8,32]
9	$[-x,\, -x+y,\, -z]$	[9,33]
10	$[x,\, x-y,\, -z]$	[10,34]
11	$[-x+y,\, y,\, -z]$	[11,35]
12	$[-y,\, -x,\, -z]$	[12,36]
13	$[-x,\, -y,\, -z]$	[13,37]
14	$[-x+y,\, -x,\, -z]$	[14,38]
15	$[y,\, -x+y,\, -z]$	[15,39]
16	$[x,\, y,\, -z]$	[16,40]
17	$[x-y,\, x,\, -z]$	[17,41]
18	$[-y,\, x-y,\, -z]$	[18,42]
19	$[-x+y,\, y,\, z]$	[19,43]
20	$[-y,\, -x,\, z]$	[20,44]
21	$[x,\, x-y,\, z]$	[21,45]
22	$[-x,\, -x+y,\, z]$	[22,46]
23	$[x-y,\, -y,\, z]$	[23,47]
24	$[y,\, x,\, z]$	[24,48]